

Project Ray

Testing Experience-Based Knowledge Accumulation in a Humanoid AI Agent

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Project description

This project explores whether a dialogue-based AI agent can simulate aspects of childlike development through repeated interaction, memory, and contextual adaptation.

Ray is a humanoid robot that learns from experience and interaction instead of starting with pre-loaded personal memories. Over time, it should personalize and adapt through what it experiences. It starts with minimal knowledge and grows through conversations and real-world input.

The system receives speech input from the user and visual input from a camera. Its knowledge base begins as a minimal vocabulary and gradually expands through stored memories from past interactions. The output is speech and movement.

For example if the context is knowing a cat, it sees a cat through the camera in real time and hears the user saying 'this is a cat and it is fluffy' and memorizes it. Also stores visual input of what a cat looks like. Retrieves description and definition of words from the knowledge base as 'white' or 'fluffy' to understand it. It should have minimal information just enough to be able to gain more. A minimal knowledge base grows through natural language interaction. The default response is asking questions.

I want it to have a curious personality suitable for learning new things and imitating human behavior and growth, but I don't want to prompt it to be that way. I want to test whether it can build a certain personality and whether a personality can emerge from architecture alone, without being prompted into existence.

What I have tried

The first version used RAG with Qwen2.5-1.5B, FAISS for vector search, and the Google Synthetic-Persona-Chat dataset as the memory source. The retrieval worked.

The problem was the dataset. PersonaChat gave Ray 45,000 conversation turns it never had. It started with memories of things it never experienced. That directly contradicts what the project is supposed to be. The mechanism was fine. The source was wrong.

The second problem was using a system prompt to define Ray's behavior. Anything defined in a prompt is static. It does not change based on what Ray experiences. If the behavior is written in advance, there is no room for it to develop from interaction. That is the opposite of what this project is trying to do.

What I want instead

Ray starts with no pre-loaded conversations and no behavioral instructions. The only starting point is a small vocabulary of basic word definitions. That is the minimal base that gives Ray enough to understand language without already knowing the world.

As Ray interacts, the memory grows. Retrieved memories become the context for generating responses. The more Ray experiences, the more its responses reflect those experiences. Nothing about its behavior is defined in advance. It develops through what it encounters.

The core research question is whether a memory system on top of a frozen language model can produce genuine knowledge accumulation through interaction, rather than just imitating knowledge that was loaded at training time.

What I am thinking of trying

Whisper – pretrained hearing tool from OpenAI. Converts audio to text.

CLIP Converts a camera frame into a vector. (pretrained, OpenAI)

Qwen2.5-1.5B

FAISS

Retrieval-Augmented Generation + Developmental architecture + confidence threshold rule

Questions:

1. Is implementation direction correct?
2. Ray is supposed to learn only from experience, but the base model already knows things. When Ray retrieves a memory, how do I make sure it uses that memory instead of what the model already knows internally? Or is that not really controllable?
3. How small can the starting vocabulary realistically be? I want the minimum that makes the system functional without giving Ray too much. Where is that line?
4. How does Ray decide what is worth storing? If someone says I saw an interesting bird today, is bird a new concept to store or just context? What makes a concept worth storing?
5. As Ray experiences more things, its memory grows. I'm worried that early important memories get buried under newer ones and become hard to retrieve. How do I keep important early experiences accessible?
6. If Ray learns what a cat is, and later learns what a kitten is, should the system explicitly record that they are related, or will storing them both as vectors naturally make Ray connect them when it needs to?
7. I want Ray's memories to get stronger the more something is repeated, and to update when the user corrects it. Is giving each memory a simple score that goes up with repetition a good enough approach, or is there a better way to handle this?

